Zaima Zarnaz

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Education

University of South Florida Jan. 2024 – Present
Ph.D. in Computer Science and Engineering Tampa, FL

- **Research Areas:** Computer Vision, Multimodal Learning, Saliency, Medical AI
- **Lab:** CSAIL Lab, advised by Dr. Seungbae Kim

Khulna University of Engineering & Technology Jan. 2017 – Apr. 2022
B.Sc. in Computer Science and Engineering Khulna, Bangladesh

Research Experience

Graduate Researcher, CSAIL Lab Jan. 2024 – Present
University of South Florida Tampa, FL

- Conduct research in computer vision, multimodal learning, saliency prediction, and medical AI.
- Develop deep learning models for visual understanding tasks using image, video, and multimodal data.
- Implement model training and evaluation pipelines using Python and PyTorch.

Selected Publications

- Object-Grounded Text Priors for Saliency Ranking Detection. *Submitted to WACV*.
- Bias in Quantized MLLMs: Abstention Shifts and Output Instability. *KDD 2026 Workshops*.
- The Interspeech 2024 TAUKADIAL Challenge: Multilingual Mild Cognitive Impairment Detection with Multimodal Approach. *Interspeech, 2024*.
- Climate Change Myths Detection Using Dynamically Weighted Ensemble-Based Stance Classifier. *ICCA, 2022*.
- Progressive Recommendation by Incremental Tensor Factorization. *ICCIT, 2022*.
- Credit Card Approval Prediction by Non-Negative Tensor Factorization. *ICREST, 2021*.

Work Experience

REVE Systems Jun. 2023 – Nov. 2023
Senior Software Engineer Dhaka, Bangladesh

- **REVE Conference:** Low-bandwidth video conferencing platform.
  - Developed and maintained the Android application.
  - Built and managed the screen-sharing module.
  - Contributed to client-server audio/video transmission protocols.
- **ICX Media:** Core media infrastructure for REVE Systems' IP telephony applications.
  - Improved system scalability from approximately 10,000 to over 40,000 concurrent calls.

Projects

Transformer-Augmented Multi-Agent Reinforcement Learning 2026

- Explored transformer-based visual memory modeling for reinforcement learning agents in a competitive multi-agent environment.
- Integrated transformer augmentation to improve long-range observation processing and agent decision-making.

Deepfake Voice Detector 2025

- Developed a deep learning-based voice deepfake detection model for distinguishing real and synthetic speech.
- Implemented model training and evaluation pipelines to assess detection performance on audio data.

Automated Revolving Camera 2019

- An automated camera system using facial recognition to track a selected face.
- Integrated a microprocessor-controlled rotation mechanism to adjust camera direction based on face movement.

Skills

Programming: Python, Java, C++, SQL, HTML/CSS
ML/CV: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, OpenCV, saliency, face recognition, multimodal learning
Tools: Docker, GitHub, Linux, Android development, Jira, PostgreSQL, MySQL